

Optimal Trading Filters: a Wiener–Hopf Approach

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Abstract

A trader holding a return-predicting signal must weigh the gain a forecast promises against the cost of realizing it and the risk of holding it, acting on current information alone. The problem is a quadratic optimization: the gain of the position against the return, net of three frictions — temporary impact, transient impact through a propagator kernel, and mean–variance inventory risk. The friction operator is non-local in time, so the unconstrained optimum needs the signal’s future, and adaptedness is the binding constraint. Existing treatments characterize the adapted optimum implicitly, as the solution of an integral equation or an equivalent forward–backward system. Using a Wiener–Hopf factorization of the friction into causal and anticausal parts, we compute it in closed form away from horizon boundaries, by whitening the forecast, projecting onto the past, and recolouring. For a stationary Gaussian signal this is an explicit linear *trading filter*. It reduces to the Markowitz rule under risk alone and, under pure power-law impact, to a fractional derivative of the signal of order $\nu = (1-\beta)/2$ with β the impact exponent; rational frictions give filters with finitely many exponential moving averages, recovering the Gârleanu–Pedersen aim portfolio and the stationary Neuman–Voß solution. The causal factor on the imaginary axis yields a closed-form value of information, a causality gap $\sin(\pi\beta/2)$ under power-law impact, and a rate response that identifies when the optimum trades against its own transient impact. On a finite horizon the factors become terminal-anchored Gohberg–Krein factors, recovering the Gatheral–Schied–Slynko and Forde–Sánchez-Betancourt–Smith solutions. All closed forms are checked against discretized adapted optima.

1 Introduction

1.1 The gain–risk–cost problem

Optimal execution studies how to build or unwind a position when trading moves the price against the trader (Almgren and Chriss 2001, Gatheral et al. 2012). A return-predicting signal sharpens the problem: a forecast invites a position, but the trade that captures it pushes the price away, commits capital to risk, and takes time the forecast may not survive. The trader weighs the gain a signal promises against the cost of realizing it and the risk of holding it, acting on the information in hand and never on the signal’s future.

Concretely, a trader observes a return signal μ_t , the conditional expectation at time t of the asset’s instantaneous price drift, and holds a position x_t , traded at the rate $u_t = \dot{x}_t$. Trading

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incurs friction in three ways. An inventory cost penalizes holding large positions. A temporary impact cost, quadratic in the rate, prices the demand for immediate liquidity. A transient impact cost carries the memory that each trade moves the price against later trades, decaying through a propagator kernel g (Bouchaud et al. 2004, Gatheral 2010).

Almgren and Chriss (2001) split the cost of trading in two: a temporary part, felt only by the trade that causes it, and a permanent part, linear in cumulative volume and left behind for every later trade. Minimizing the resulting mean–variance functional produced closed-form execution schedules. Empirical work then dissolved the sharp split: Lillo et al. (2003) found impact to be a concave function of size, and Bouchaud et al. (2004) found it *transient*, a trade displacing the price and the displacement relaxing, so that permanent and temporary costs are the slow and fast limits of one decaying response. This is the propagator picture, in which the price carries a weighted memory of past trades through a kernel g , and the shape of g is the modeling choice.

The tractable shape is *exponential*: Obizhaeva and Wang (2013) derived $g(t) = e^{-\kappa t}$ from a limit order book of finite depth refilling at a fixed rate, a single resilience time $1/\kappa$ whose fast and slow limits return the temporary and permanent costs of Almgren and Chriss (2001), and it underlies much of the closed-form execution literature. Transaction data across markets then made clear that the decay is instead a *power law*, $g(t) = t^{-\beta}$ with $\beta \in (0.2, 0.6)$ (Bouchaud et al. 2004, Jusselin and Rosenbaum 2020), scale-free and long-memory; Gatheral (2010) showed a decay of about this form to be what no-dynamic-arbitrage allows, and Jusselin and Rosenbaum (2020) tied the exponent to rough volatility. The exponential kernel offers finite memory and closed forms; the power law has long memory and the empirical evidence.

The trader maximizes the expected gain of the position against the return, net of these costs,

$$\max_{x \text{ adapted}} \mathbb{E} \int x_t \mu_t dt - \frac{\eta}{2} \mathbb{E} \int \dot{x}_t^2 dt - \frac{\gamma}{2} \mathbb{E} \iint g(|t-s|) \dot{x}_t \dot{x}_s dt ds - \frac{\lambda}{2} \mathbb{E} \int x_t^2 dt, \quad (1)$$

over positions adapted to the flow of information: x_t may depend on the signal observed up to time t , not on its future. The gain pairs the position against the expected return, the mean–variance objective of portfolio choice; the three costs are the frictions of execution. The integrals run over the whole line; Section 4 poses the same problem on a finite window $[0, T]$ with a terminal inventory constraint. This paper gives exact solutions to (1) under both exponential and power-law impact kernels.

1.2 The adaptedness constraint

The objective (1) pairs a linear gain $\mathbb{E} \int x_t \mu_t dt$ against a positive quadratic cost $\frac{1}{2} \mathbb{E} \langle x, Nx \rangle$ and thus evaluates the Legendre transform of the cost at the signal, under the adaptedness constraint. Referred to the position, the three costs define a single *friction operator* $N = -\eta \partial_t^2 - \gamma \partial_t^2(g*) + \lambda I$, with Fourier symbol

$$\hat{n}(\omega) = \eta \omega^2 + \gamma \hat{g}(\omega) \omega^2 + \lambda, \quad \hat{g}(\omega) = \int e^{i\omega t} g(|t|) dt \geq 0. \quad (2)$$

Referred to the trading rate the friction is $Q = N (-\partial_t^2)^{-1}$, with symbol $\hat{q} = \hat{n}/\omega^2 = \eta + \gamma \hat{g} + \lambda/\omega^2$, the form used for the scale-free kernel of Section 3. Could the trader see the whole signal path, the first-order condition of (1) would be a single operator inversion, the position

$$x = N^{-1} \mu, \quad (3)$$

the continuous-time, memory-laden analogue of the Markowitz rule $w = \lambda^{-1} \Sigma^{-1} \mu$ (weights w , covariance Σ , expected returns μ), with the friction operator in place of the covariance. But the signal path is revealed only as time passes, and (3) is acausal.

With inventory cost alone ($\eta = \gamma = 0$) the quadratic form is diagonal in time: the first-order condition at time t involves x_t and μ_t only, and the adapted optimum is the Markowitz rule $x_t = \mu_t/\lambda$ (Section 5). A temporary cost adds the local operator $-\eta \partial_t^2$, still an ordinary differential equation. Transient impact removes this locality: its symbol $\gamma \hat{g}(\omega) \omega^2$ is not a polynomial, so Nx at time t depends on the position over the past *and the future*, and adaptedness — the requirement that x_t be measurable with respect to $\mathcal{F}_t$ — becomes binding. Varying (1) over adapted perturbations gives the *projected* first-order condition,

$$\mathbb{E}_t[(Nx^*)(t)] = \mu_t \quad \Longleftrightarrow \quad P_+ N P_+ x^* = \mu, \quad (P_+ X)_s = \mathbb{E}_s[X_s], \quad (4)$$

with P_+ the optional projection onto adapted processes. Solving (4) requires inverting $P_+ N P_+$, and the inverse of a projected operator is not the projection of the inverse, precisely because N is non-local.

1.3 The Wiener–Hopf method

Wiener and Hopf (1931) introduced the technique that resolves this class of problems, originally for convolution equations on a half-line, $\int_0^\infty k(t-s)f(s)ds = g(t)$ for $t \geq 0$ (Krein 1962, Noble 1958). Fourier analysis diagonalizes convolutions on the whole line; a half-line support constraint couples the frequencies, in the same way the filtration constraint couples them in (4). The method restores solvability by factoring the symbol,

$$\hat{k}(\omega) = \hat{k}_-(\omega) \hat{k}_+(\omega),$$

with $\hat{k}_+$ analytic and zero-free in the upper half-plane and $\hat{k}_-$ in the lower. Analyticity in a half-plane encodes causality: by Paley–Wiener, $\hat{k}_+$ is the transform of a kernel supported on $t \geq 0$ and $\hat{k}_-$ of one supported on $t \leq 0$. Operators with causal kernels are triangular with respect to the time order, and triangular operators invert forward in time, staying triangular. The factorization thus splits a two-sided convolution into two one-sided operators, each compatible with the boundary, between which the half-line projection can be threaded. The finite-dimensional analogue is the Cholesky factorization of a positive matrix into triangular factors, computed by Gram–Schmidt along the coordinate order.

The method’s classical uses are prediction and control. For linear prediction of a stationary time series the factored symbol is the spectral density, $S = |\hat{\psi}|^2$, and the causal factor builds the innovations representation (Wiener 1949, Whittle 1963). In the trading problem the factored symbol is the *friction* N , and the causal factor builds the optimal trade: prediction whitens by the signal’s own factor to forecast it; trading whitens by the friction’s factor to monetize it. Existence of such a factorization for a general positive operator was established later — Gohberg and Krein (1970) for Volterra operators on an interval, Arveson (1975) for positive operators on a nest of subspaces — and these existence theorems do not by themselves produce the factor. It is explicit in the stationary case, where Fourier diagonalizes N and the causal factor is the scalar Szegő outer function; frequency-domain factorization of this kind also solves linear–quadratic dynamic optimization in economics (Whiteman 1985). On a finite horizon translation invariance is lost and the factor is explicit only for the special kernels of Section 4.

Section 2 solves the stationary problem in three steps — factor the friction, predict the signal from its past, and combine the two — through a projected-inverse identity (Lemma 1) that assembles the adapted optimum for a general signal (Theorem 1) and, for stationary Gaussian signals, an explicit trading filter (Theorem 2). The remaining sections draw out the consequences. Section 3 collects the peculiarities of pure power-law impact: a fractional-derivative policy, a scale-free

value of anticipation, and a trade that never reverses against the signal. Section 4 treats the finite horizon, and Section 5 recovers the earlier portfolio and execution solutions as special cases of the factorization.

1.4 Relation to earlier work

The execution literature has treated pieces of (1) by several routes: deterministic liquidation under general propagators (Gatheral et al. 2012); signal-adaptive trading, introduced for Almgren–Chriss costs by Cartea and Jaimungal (2013) and developed under temporary and exponential-resilience costs (Lehalle and Neuman 2019, Neuman and Voß 2022, Bank et al. 2017); general propagators through infinite-dimensional stochastic control (Abi Jaber and Neuman 2025) and operator-resolvent methods (Abi Jaber et al. 2024b); and power-law resilience with Gaussian signals on a finite horizon (Forde et al. 2022). Each computes an inverse of the same friction operator. The finite-horizon methods produce that inverse as a resolvent series, a Wiener-chaos expansion, or the solution of a Riccati system; the factorization sums those series in closed form when translation invariance is available. Path-dependent and data-driven strategies — signature methods (Kalsi et al. 2020) and reinforcement learning — approximate the problem and lie outside the closed-form scope here.

The two representations differ in what they expose. Dynamic programming and the backward-SDE methods return a *feedback law*: the trade is a function of the current state — the position, the outstanding transient-impact profile, and the signal — with gains fixed by a Riccati or backward SDE, and its dependence on the signal is implicit, carried by the state dynamics one propagates forward. The factorization returns a *filter*: the trade is an explicit linear operator on the signal, a transfer function mapping the observed signal to the trade in closed form — a gain and phase at each frequency, a fractional derivative for power-law impact, a few moving averages for a rational friction.

Abi Jaber and Neuman (2025) carry the feedback route to general propagators, characterizing the optimum through a free-boundary backward SDE and an operator Riccati equation, with existence and uniqueness for any Volterra propagator and any adapted signal and closed forms recovered by kernel-specific reductions; the variational form of the same problem is a stochastic Fredholm equation, extended to linear constraints by Lagrange multipliers in Abi Jaber et al. (2024a). The present approach trades that generality for explicitness: restricted to the stationary regime, it expresses the adapted inverse of the friction directly through the factorization, as a closed-form filter. The two describe the same adapted optimum; Section 5.3 makes the identification, the iterative solution of the integral equation appearing as the Neumann expansion that the Gohberg–Krein factors sum in closed form.

2 The stationary solution

The stationary problem is solved in the three steps announced in §1.3: factor the friction (§2.1), predict the signal (§2.2), and combine the two into a filter (§2.3). Throughout, fix a filtered probability space satisfying the usual conditions; Table 1 collects the notation. The return signal μ is adapted and mean-zero, with forecast curve

$$\bar{\mu}(t, s) = \begin{cases} \mu_s, & s \leq t, \\ \mathbb{E}_t[\mu_s], & s > t, \end{cases}$$

Symbol	Meaning
<i>Operators</i> (act on time-domain processes; no argument)	
$N, N_{\pm}$	friction operator (position-referred) and its causal/anticausal Szegő factors
$Q, Q_{\pm}$	friction referred to the rate, $Q = N(-\partial_t^2)^{-1}$, and its factors (§3)
P_+	optional projection onto adapted processes
$I_{\pm}^{\nu}, D_{\pm}^{\nu}$	Liouville fractional integrals and Marchaud fractional derivatives
$C_{\pm}$	Gohberg–Krein (finite-horizon Volterra) factors
G_T, I	rate-referred friction on $[0, T]$; identity
<i>Kernels and frequency-domain symbols</i> ($\hat{\cdot}$ denotes the Fourier transform)	
g	transient-impact propagator kernel
$\hat{n}$	friction symbol $\eta\omega^2 + \gamma\hat{g}\omega^2 + \lambda$; outer factors $\hat{n}_{\pm}$
$\hat{q} = \hat{n}/\omega^2$	rate-referred symbol $\eta + \gamma\hat{g} + \lambda/\omega^2$, with $\hat{q}_{\pm} = \hat{n}_{\pm}/(\mp i\omega)$
$\hat{\psi}, \hat{\phi}$	outer spectral factors of the return μ ($S_{\mu} = \hat{\psi} ^2$) and of the appreciation α (§3)
$[\cdot]_+$	causal (plus) part of a symbol
<i>Filters and processes</i>	
$\hat{\pi}, \hat{\chi} = (-i\omega)\hat{\pi}$	position filter and trade (rate) filter: innovations $\dot{W} \mapsto x^*, u^*$
μ, α	return signal (expected return) and expected remaining appreciation $\alpha_t = \mathbb{E}_t \int_t^{\infty} \mu_s ds$
u, x	trading rate and position, $x_t = \int_{-\infty}^t u_s ds$; forecast curve $\bar{\mu}(t, s) = \mathbb{E}_t[\mu_s]$
$W, \dot{W}$	Brownian motion and its innovations (white noise)
ζ, ξ^*	whitened forecast $N_-^{-1}\bar{\mu}$; nonanticipativity multiplier
<i>Scalars</i>	
η, γ, λ	temporary-impact weight, transient-impact weight, risk aversion
θ, κ	signal mean-reversion rate; impact resilience rate
β, ν	propagator exponent; fractional order $\nu = (1 - \beta)/2$
c_{β}	power-law transform constant, $\hat{g} = c_{\beta} \omega ^{\beta-1}$, $c_{\beta} = 2\Gamma(1 - \beta)\sin(\pi\beta/2)$
$\Phi(\theta)$	outer factor on the imaginary axis, $\hat{n}_+(i\theta)$
$R(\theta), X(\theta)$	rate response and position response
v, v_{ant}	adapted value rate; anticipative (perfect-foresight) value rate

Table 1: Notation. Operators are italic capitals and carry no argument; kernels are lowercase Latin; a hat denotes the Fourier transform; filters are lowercase Greek. The development is in the position variable; the rate variable is used for the scale-free kernel of Section 3, where the position is non-stationary, and on the finite horizon of Section 4.

decaying as $s \rightarrow \infty$, so that forecasts vanish at long horizons. The Fourier convention is $\hat{f}(\omega) = \int e^{i\omega t} f(t) dt$, under which causal kernels have transforms analytic in the upper half-plane. Admissible positions are adapted processes of finite friction energy $\mathbb{E}\langle x, Nx \rangle < \infty$; for $\eta = 0$ this class admits positions with a Brownian component, whose trading rate carries white noise.

2.1 Factorization of the friction

The symbol $\hat{n}$ of (2) is positive and even, grows polynomially, and satisfies the Szegő condition $\int |\log \hat{n}(\omega)|/(1 + \omega^2) d\omega < \infty$, met by every kernel considered here, the pure power law included: its zero $\hat{n} \sim \gamma c_{\beta} |\omega|^{1+\beta}$ at $\omega = 0$ is only logarithmic in $\log \hat{n}$, hence integrable (Krein 1962). This log-integrability lets N factor as $N = N_- N_+$ with symbol

$$\hat{n}(\omega) = \hat{n}_-(\omega) \hat{n}_+(\omega), \quad \hat{n}_- = \overline{\hat{n}_+}, \quad (5)$$

where $\hat{n}_+$ is analytic and zero-free in the upper half-plane. Then N_+ is causal with causal inverse, and $N_- = N_+^*$ is its anticausal adjoint. Conjugation by such a factor whitens the friction: under

it the inner product $\langle \cdot, N \cdot \rangle$ becomes the flat L^2 product. The rate-referred factors carry one time-integration on each side, $\hat{q}_\pm = \hat{n}_\pm / (\mp i\omega)$. The outer factor $\hat{n}_+$ is given by the Szegő formula

$$\hat{n}_+(\omega) = \exp\left(\frac{1}{2\pi i} \int_{-\infty}^{\infty} \left(\frac{1}{t-\omega} - \frac{t}{1+t^2}\right) \log \hat{n}(t) dt\right), \quad \text{Im } \omega > 0, \quad (6)$$

unique up to a unimodular constant (Krein 1962). Because $\hat{n}$ is even, evaluating (6) on the imaginary axis reduces to the real Poisson integral $\Phi(\theta) := \hat{n}_+(i\theta)$ of (14), used repeatedly below.

Assumption 1 (Friction). The weights satisfy $\eta, \gamma, \lambda \geq 0$; the propagator g is locally integrable and positive-definite, $\hat{g} \geq 0$ in (2); and the symbol $\hat{n}$ is positive almost everywhere and satisfies the Szegő condition, so that the factorization (5) exists.

2.2 Prediction of the signal

The adapted trade responds to the signal's expected future. The trader cannot see the future path of μ and replaces it by the forecast curve $\bar{\mu}(t, s) = \mathbb{E}_t[\mu_s]$, $s \geq t$. For a Gaussian signal the forecast is a linear functional of the past; it reduces to a linear operator on the signal itself, or on a finite Markov lift, when the signal is finite-dimensional.

Definition 1 (Markov signal). A stationary signal is Markov of order d if its forecast curve is a deterministic combination of the current value and its first $d - 1$ derivatives,

$$\bar{\mu}(t, s) = \sum_{k=0}^{d-1} \mu_t^{(k)} g_k(s-t) \quad (s \geq t), \quad g_k(0) = \delta_{k0}, \quad (7)$$

equivalently if its spectral density is rational of degree d .

Such a signal is a linear readout $\mu_t = c^\top X_t$ of a d -dimensional Markov state — its lift; in observable coordinates the lift is $(\mu_t, \dot{\mu}_t, \dots, \mu_t^{(d-1)})$, and (7) carries the whole future in finitely many numbers. Consistency pins the profiles down: the tower property $\mathbb{E}_t[\bar{\mu}(t', s)] = \bar{\mu}(t, s)$ forces the mean dynamics to be linear, so the g_k solve a d th-order constant-coefficient ODE — exponential modes $e^{-r_j(s-t)}$, or damped oscillations when the rates r_j are complex.

The mean-reverting case is $d = 1$: the forecast is the current value times one profile, $\bar{\mu}(t, s) = \mu_t f(s-t)$, and the tower property forces $f(\tau) = e^{-\theta\tau}$ with $\theta > 0$ the mean-reversion rate. This is the Ornstein–Uhlenbeck signal — the only stationary Gaussian signal with a fixed forecast profile, by Doob's theorem — and it also covers the non-Markov, non-Gaussian moving averages $\mu_t = \int_{-\infty}^t e^{-\theta(t-r)} \sigma_r dW_r$ with adapted stochastic volatility σ . For $d = 2$ the forecast combines the signal and its velocity, a damped oscillation when the two rates are complex.

A signal with no finite-dimensional state has no such reduction: its forecast is the Wiener–Kolmogorov predictor, a linear operator on the whole past. Take μ stationary, Gaussian, purely non-deterministic, and generating its own filtration $\mathcal{F}_t = \sigma(\mu_s : s \leq t)$. Its spectral density factors as $S_\mu = |\hat{\psi}|^2$ with $\hat{\psi}$ the outer (causal, zero-free) spectral factor, so $\mu = \psi * \dot{W}$ is a causal moving average of its innovations $\dot{W}$, the unit white noise generating the filtration — the Wold representation (Wiener 1949, Whittle 1963). The forecast keeps the innovations up to t and drops the later ones.

2.3 The trading filter

The factorization and the forecast combine through the adapted inverse of N . The first-order condition (4) inverts the projected operator $P_+ N P_+$, which the one-sided factors assemble. The

lemma is stated for a general factored positive operator; it is applied to N here, to the rate-referred Q in Section 3, and to the finite-horizon operator in Section 4.

Lemma 1 (Adapted projected inverse). *Let $A = A_- A_+$ be a positive operator with A_+ causal with causal inverse and $A_- = A_+^*$ anticausal with anticausal inverse. Then, on adapted processes,*

$$(P_+ A P_+)^{-1} = A_+^{-1} P_+ A_-^{-1}. \quad (8)$$

The proof, in Appendix A, is triangular bookkeeping: causality makes A_+ and its inverse preserve the adapted subspace, adjointness makes A_- and its inverse preserve the orthogonal complement, and the projection annihilates what lands there. Among all square roots of $\hat{n}$, the factorization (5) selects the one whose whitening N_-^{-1} leaves the space of processes invisible to adapted strategies invariant; this is the sense in which it is Cholesky with respect to the filtration.

Remark 1 (Relation to operator factorization). Lemma 1 is the projected-inverse identity behind the triangular factorization of a positive operator relative to a nest of subspaces (Gohberg and Krein 1970, Arveson 1975). Here the nest is the increasing family of adapted subspaces indexed by the filtration, and (5) is the corresponding filtration-indexed Cholesky factorization: the scalar Szegő outer factor in the stationary case, and the Gohberg–Krein Volterra factor on a finite horizon (Section 4).

Theorem 1 (Optimal policy, general adapted signal). *Under Assumption 1, the unique adapted optimum of (1) is*

$$x^* = N_+^{-1} P_+ N_-^{-1} \mu, \quad \text{equivalently} \quad x_t^* = (N_+^{-1} \zeta)(t), \quad \zeta_s = (N_-^{-1} \bar{\mu}(s, \cdot))(s), \quad (9)$$

where the second form uses the commutation of the deterministic operator N_-^{-1} with conditional expectation on its argument.

The three factors in (9) are the three steps of the classical recipe: an anticausal whitening, the adapted projection, and a causal recolouring. The whitening would call for the future of the signal; the projection supplies instead its forecast (§2.2). Only the friction’s factor enters the whitening; the signal’s own spectral factor enters through the projection alone. Completing the square gives a second reading. Up to a constant the objective (1) equals $-\frac{1}{2} \|x - N^{-1} \mu\|_N^2$ in the friction norm $\|y\|_N^2 = \mathbb{E}\langle y, Ny \rangle$, so that x^* is the orthogonal projection, in that norm, of the anticipative optimum $N^{-1} \mu$ onto the adapted positions — equivalently the causal Wiener–Kolmogorov estimate of the whitened return from the past.

Remark 2 (Relation to Wiener–Kolmogorov prediction). The construction is the Wiener–Hopf solution of a Wiener–Kolmogorov estimation problem, with the friction in place of the signal covariance. Linear prediction whitens a stationary signal by its own spectral factor $\hat{\psi}$, keeps the causal part, and recolours, returning the least-squares forecast from the past (Wiener 1949, Whittle 1963). The same three steps, carried out with the friction’s factor $\hat{n}_+$ in place of $\hat{\psi}$, give the optimal position: by the estimation reading above, x^* is the causal estimate of the anticipative optimum $N^{-1} \mu$ in the friction metric. Prediction and trading are thus one construction on two quadratic forms, the signal covariance and the friction — the correspondence that carries frequency-domain factorization from filtering into linear–quadratic control (Whiteman 1985).

Remark 3 (The nonanticipativity multiplier). Substituting (9) into Nx^* writes the first-order condition (4) without the projection,

$$N x^* = \mu - \xi^*, \quad \xi^* = N_- (I - P_+) N_-^{-1} \mu, \quad P_+ \xi^* = 0, \quad (10)$$

so the adapted optimum solves the unconstrained first-order condition for the modified return $\mu - \xi^*$. A multiplier with vanishing optional projection is the defining property of the nonanticipativity multipliers of Rockafellar and Wets (1976), established there for general convex multistage problems; (10) is their closed form here. Because $\mathbb{E}\langle x, \xi^* \rangle = 0$ for adapted x , the value forgone to adaptedness equals the friction energy $\frac{1}{2}\mathbb{E}\langle \xi^*, N^{-1}\xi^* \rangle$ of the multiplier, per unit time in the stationary case — the shadow price of information of §2.4 below.

Assumption 2 (Signal). The return signal μ is stationary, Gaussian, purely non-deterministic, and generates its own filtration, with Wold representation $\mu = \psi * \dot{W}$ and spectral density $S_\mu = |\hat{\psi}|^2$, $\hat{\psi}$ the outer spectral factor.

For a stationary signal the projection is itself a filter. In the innovations $\dot{W}$ the whitened return has symbol $h := \hat{\psi} \hat{n}_-^{-1}$; since conditioning at time s discards the innovations after s , the projection truncates the moving-average kernel at lag zero and acts on symbols as the Riesz plus-part $[\cdot]_+$ onto non-negative lags. Then (9) is a transfer function.

Theorem 2 (Optimal trading filter). *Under Assumptions 1 and 2, and $\lambda > 0$, the optimal position is the linear filter $x^* = \pi * \dot{W}$ with transfer function*

$$\hat{\pi} = \hat{n}_+^{-1} [h]_+, \quad h = \hat{\psi} \hat{n}_-^{-1}, \quad (11)$$

the optimal rate the filter $\hat{\chi} = (-i\omega) \hat{\pi}$, and the attained value rate, the expected profit per unit time, is

$$v = \frac{1}{4\pi} \|[h]_+\|_{L^2}^2. \quad (12)$$

For a Markov signal the same filter reads on the observed signal rather than the innovations. The whitened forecast then reduces to a differential operator: with $\rho_k = (N_-^{-1} g_k)(0)$,

$$\zeta_s = (N_-^{-1} \bar{\mu}(s, \cdot))(s) = \sum_{k=0}^{d-1} \rho_k \mu_s^{(k)} = P(\partial) \mu_s, \quad (13)$$

so that the policy filters the signal and its derivatives, $\hat{x}^*(\omega) = P(-i\omega) \hat{\mu}(\omega) / \hat{n}_+(\omega)$, a rational filter. For the Ornstein–Uhlenbeck signal ($d = 1$) the single pole at $\omega = -i\theta$ gives $P(\partial) = \rho = 1/\hat{n}_-(-i\theta) = 1/\Phi(\theta)$ (Appendix B), with

$$\Phi(\theta) := \hat{n}_+(i\theta) = \exp\left[\frac{\theta}{2\pi} \int \frac{\log \hat{n}(t)}{\theta^2 + t^2} dt\right] > 0, \quad (14)$$

the outer factor of $\hat{n}$ on the imaginary axis, so the position filter is the single formula covering all three frictions,

$$\hat{x}^*(\omega) = H(\omega) \hat{\mu}(\omega), \quad H(\omega) = \frac{1}{\Phi(\theta) \hat{n}_+(\omega)}, \quad (15)$$

with value $v = \sigma^2 \theta / 4\Phi(\theta)^2$, where $\theta^2 \sigma^2$ is the innovation variance of μ (σ^2 that of $\alpha = \mu/\theta$). For $d = 2$ the filter uses the signal and its velocity.

The hypothesis $\lambda > 0$ keeps the whitened return and the position filter in L^2 : without risk aversion $\hat{n}$ vanishes at $\omega = 0$ and the position is non-stationary — the regime of Section 3, where the theory is referred to the rate. The rate filter $\hat{\chi}$ lies in L^2 under the spectral-decay hypothesis of Appendix B, and otherwise, notably for the exponential kernel without temporary cost, carries a lag-zero white-noise atom (Section 3.2).

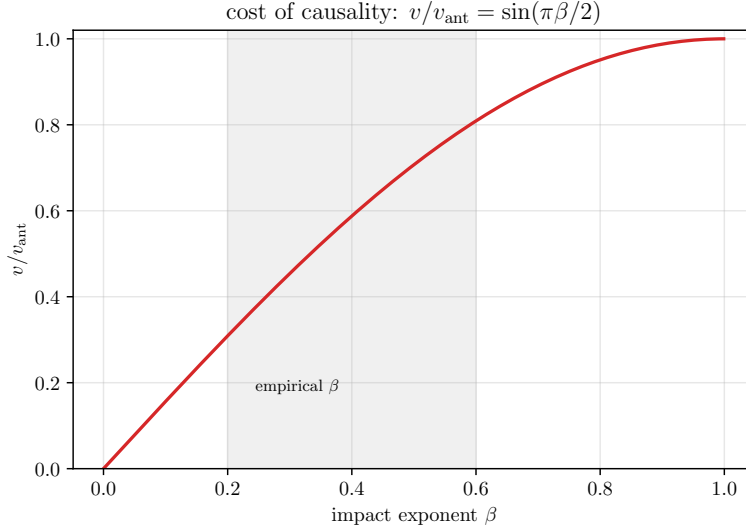


Figure 1: The cost of causality. Under power-law impact the adapted value is a fixed fraction of the clairvoyant value, $v/v_{\text{ant}} = \sin(\pi\beta/2)$ of (17), independent of the signal speed: adaptedness is nearly free for short-memory impact ($\beta \rightarrow 1$) and forfeits almost all the value for near-permanent impact ($\beta \rightarrow 0$), keeping between a third and four-fifths of the clairvoyant value over the empirical range (shaded).

2.4 The value of information

The value (12) has a benchmark. The perfectly-anticipating trader replaces the projection $[\cdot]_+$ by the identity, extracting

$$v_{\text{ant}} = \frac{1}{4\pi} \int \frac{S_\mu}{\hat{n}} d\omega, \quad (16)$$

the value attainable with the whole signal path in hand. Equation (16) is the position-referred form, finite for $\lambda > 0$; for the scale-free kernel at $\lambda = 0$ the position form diverges, and the anticipative value is referred to the rate, $v_{\text{ant}} = \frac{1}{4\pi} \int S_\alpha/\hat{q} d\omega$, finite there and used in Appendix B to obtain (17). The *causality gap* is the ratio $v/v_{\text{ant}} \leq 1$, the fraction of the clairvoyant value that survives the restriction to current information. The shortfall $v_{\text{ant}} - v$ equals, by (12), the L^2 energy of the anticausal part of the half-whitened signal $h = \hat{\psi}\hat{n}_-^{-1}$, and by Remark 3 the friction energy of the nonanticipativity multiplier ξ^* — the shadow price of adaptedness, in closed form. For an Ornstein–Uhlenbeck signal the value $v = \sigma^2\theta/4\Phi(\theta)^2$ turns on the friction factor $\Phi(\theta) = \hat{n}_+(i\theta)$.

Under pure power-law impact the causality gap is scale-free, independent of the signal speed (§3.1):

$$\frac{v}{v_{\text{ant}}} = \sin \frac{\pi\beta}{2}. \quad (17)$$

Adaptedness is nearly free for short-memory impact ($\beta \rightarrow 1$) and forfeits almost all the value for near-permanent impact ($\beta \rightarrow 0$); at the empirical $\beta \in (0.2, 0.6)$ the adapted trader keeps between a third and four-fifths of the clairvoyant value (Figure 1).

3 Pure power-law impact

The power-law kernel is scale-free: its transform $\hat{g}(\omega) = c_\beta |\omega|^{\beta-1}$ diverges as $\omega \rightarrow 0$, so the impact is not integrable and has no characteristic timescale. It is a superposition of exponentials over all decay rates, $|t|^{-\beta} = \Gamma(\beta)^{-1} \int_0^\infty e^{-r|t|} r^{\beta-1} dr$: a continuum where a rational kernel has finitely many, and the reason the optimal filter is fractional. This is long memory, in contrast with the exponential kernel $e^{-\kappa|t|}$, whose impact is integrable and dies on the single scale $1/\kappa$. The optimum reflects this memory in three ways: the policy is a fractional derivative (§3.1), the value of anticipation obeys the scale-free law (17), and the trade never reverses against the signal (§3.2).

With $\eta = \lambda = 0$ the friction symbol $\hat{n} = \gamma c_\beta |\omega|^{1+\beta}$ vanishes at $\omega = 0$: the optimal position is non-stationary, and the stationary object is the rate. This section therefore works in the rate variable and the appreciation signal $\alpha_t = \mathbb{E}_t \int_t^\infty \mu_s ds$ of §1.1, with forecast curve $\bar{\alpha}(t, s) = \mathbb{E}_t[\alpha_s] = \int_s^\infty \bar{\mu}(t, r) dr$; for the Ornstein–Uhlenbeck signal $\alpha = \mu/\theta$. Referred to the rate the friction is Q with symbol $\hat{q} = \hat{n}/\omega^2 = \gamma \hat{g}$ and factors $\hat{q}_\pm = \hat{n}_\pm/(\mp i\omega)$; Lemma 1 applies to $Q = Q_- Q_+$ as it does to N , and the policy of Theorem 1 reads

$$u^* = Q_+^{-1} P_+ Q_-^{-1} \alpha, \quad u_t^* = (Q_+^{-1} \zeta)(t), \quad \zeta_s = (Q_-^{-1} \bar{\alpha}(s, \cdot))(s), \quad (18)$$

the same three operations, with the whitened forecasts equal, $(Q_-^{-1} \bar{\alpha}(s, \cdot))(s) = (N_-^{-1} \bar{\mu}(s, \cdot))(s)$ (Appendix A). For a stationary signal the filter and value of Theorem 2 likewise refer to the rate: $\hat{\chi} = \hat{q}_+^{-1} [h_\alpha]_+$ and $v = \frac{1}{4\pi} \| [h_\alpha]_+ \|^2$ with $h_\alpha := \hat{q}_+^{-1} \hat{\varphi}$, $\hat{\varphi}$ the outer spectral factor of α ($S_\alpha = |\hat{\varphi}|^2$); these remain in L^2 at $\lambda = 0$ (Appendix B), and the causality-gap law (17) follows from the residue of the power-law factor there.

3.1 The policy is a fractional derivative

At one extreme, temporary cost alone ($\gamma = \lambda = 0$) makes the friction local: the optimal rate is the memoryless myopic rule $u_t^* = \alpha_t/\eta$, proportional to the appreciation signal α — whereas Markowitz makes the *position* proportional to the return $\mu = \mathbb{E}_t[-\dot{\alpha}]$ (Section 5). Power-law impact is the opposite extreme: its policy is a fractional derivative of the signal. The fractional integral of order $\nu \in (0, 1)$ on the whole line is the Liouville operator

$$(I_+^\nu f)(t) = \frac{1}{\Gamma(\nu)} \int_{-\infty}^t (t-s)^{\nu-1} f(s) ds,$$

with I_-^ν its time reflection; it interpolates between the identity ($\nu \rightarrow 0$) and a single integration ($\nu = 1$), weighting the past of f by the power-law kernel $(t-s)^{\nu-1}$. Its inverse $D_\pm^\nu = (I_\pm^\nu)^{-1}$, the fractional derivative of order ν , is non-local, determined by the entire history rather than the local slope; for the bounded signals here it takes the Marchaud form (Samko et al. 1993),

$$(D_\pm^\nu f)(t) = \frac{\nu}{\Gamma(1-\nu)} \int_0^\infty \frac{f(t) - f(t \mp s)}{s^{1+\nu}} ds,$$

whose increment regularizes the weight $s^{-1-\nu}$ and keeps the integral finite for bounded, non-decaying f . The power-law symbol is a fractional power of frequency: for $g(t) = |t|^{-\beta}$ the transform is $\hat{g}(\omega) = c_\beta |\omega|^{\beta-1}$, so the Wiener–Hopf factors are fractional integrals, $Q_\pm = (\gamma c_\beta)^{1/2} I_\pm^\nu$ with $\nu = (1-\beta)/2$, and their inverses are the fractional derivatives $D_\pm^\nu$. The rate policy (18) then reads

$$u_t^* = \frac{1}{\gamma c_\beta} (D_+^\nu \zeta)(t), \quad \zeta_s = (D_-^\nu \bar{\alpha}(s, \cdot))(s) : \quad (19)$$

an anticausal fractional derivative of the forecast curve, followed by a causal fractional derivative of the whitened forecast, both of order $\nu \in (0.2, 0.4)$ for the empirical range of β . For an Ornstein–Uhlenbeck signal $\zeta_s = \theta^\nu \alpha_s$ and $u^* = (\theta^\nu / \gamma c_\beta) D_+^\nu \alpha$, with value $v = \sigma^2 \theta^{-\beta} / 4 \gamma c_\beta$; comparing with the anticipative value (16) gives the causality gap (17). The filter has long memory: the Marchaud representation weights past signal increments by the slowly decaying kernel $(t-s)^{-1-\nu}$, and on a grid D_+^ν is a Toeplitz operator whose Grünwald–Letnikov weights inherit this tail.

The position that this rate accumulates is a fractional *integral* of the return. The position filter (15), $\hat{x}^* = \hat{\mu} / (\Phi(\theta) \hat{n}_+)$, has $\hat{n}_+ \propto (-i\omega)^{(1+\beta)/2}$ for the pure power law, so the position is a causal fractional integral of the return of order $(1+\beta)/2$,

$$\hat{x}^*(\omega) \propto (-i\omega)^{-(1+\beta)/2} \hat{\mu}(\omega), \quad x^* \propto I_+^{(1+\beta)/2} \mu, \quad (20)$$

the scale-free, long-memory counterpart of the memoryless Markowitz position $x^* = \mu / \lambda$. Low frequencies are amplified: the squared gain is $|\omega|^{-(1+\beta)}$, so the position has spectral density $|\omega|^{-(1+\beta)} S_\mu(\omega)$ up to a constant. Since $1+\beta > 1$, this diverges at the origin unless the return’s low-frequency power is depleted: the position is stationary if and only if $\int |\omega|^{-(1+\beta)} S_\mu(\omega) d\omega < \infty$, that is, if $S_\mu(\omega)$ vanishes at $\omega = 0$ faster than $|\omega|^\beta$. A persistent return fails the test: for the Ornstein–Uhlenbeck signal $S_\mu(0) \neq 0$, and the position drifts without bound, a fractional analogue of the random walk that integrating a persistent return produces. This is why the pure power-law theory is referred to the rate, stationary for every signal, rather than to the position; a risk term $\lambda > 0$ confines the position for any return (§2.3), at the cost of leaving the scale-free regime.

Between the two ends the position filter $H(\omega) = [\Phi(\theta) \hat{n}_+(\omega)]^{-1}$ has magnitude proportional to $\hat{n}(\omega)^{-1/2}$, interpolating across the crossover frequencies of $\hat{q}$: risk-dominated at low frequency, transient-impact-dominated in the middle, and temporary-cost-dominated at high frequency, the crossovers falling at $\omega_c = (\lambda / \gamma c_\beta)^{1/(1+\beta)}$ and $\omega_* = (\gamma c_\beta / \eta)^{1/(1-\beta)}$. Figure 2 places the two transient kernels side by side in both domains: the exponential settles on a finite-memory plateau and an impulse response that cuts off, while the power law slides without a plateau and leaves an algebraic tail, the temporary cost capping the short lag and the risk term the long.

3.2 Impact surfing

Whether the optimal trade follows the signal or runs against it is a further difference between the kernels, decided by the response of the rate rather than the position. For an Ornstein–Uhlenbeck signal define the position response $X(\theta) = \mathbb{E}[x_t^* | \alpha_t] / \alpha_t$ and the rate response $R(\theta) = \lim_{\epsilon \downarrow 0} \mathbb{E}[u_{t+\epsilon}^* | \alpha_t] / \alpha_t$; the latter is the average direction of the trade executed just after t by a trader holding the current signal value. The forward limit handles the distribution-valued rate: for the exponential kernel with no temporary cost the optimal rate carries a contemporaneous white-noise atom correlated with α_t , of weight

$$c_1 = \lim_{|\omega| \rightarrow \infty} \frac{1}{\hat{n}_+(\omega)} = \frac{1}{\sqrt{2\gamma\kappa + \lambda}} \quad (\text{exponential}), \quad c_1 = 0 \quad (\text{power-law, or any } \eta > 0); \quad (21)$$

evaluating an instant later, where the intervening innovations are independent of α_t , removes the atom and leaves the execution-relevant response.

Proposition 1 (Rate response and the surfing threshold). *For rational or power-law symbols,*

$$R(\theta) = \frac{\theta^2}{\Phi(\theta)} \left[\frac{1}{\Phi(\theta)} - 2c_1 \right], \quad (22)$$

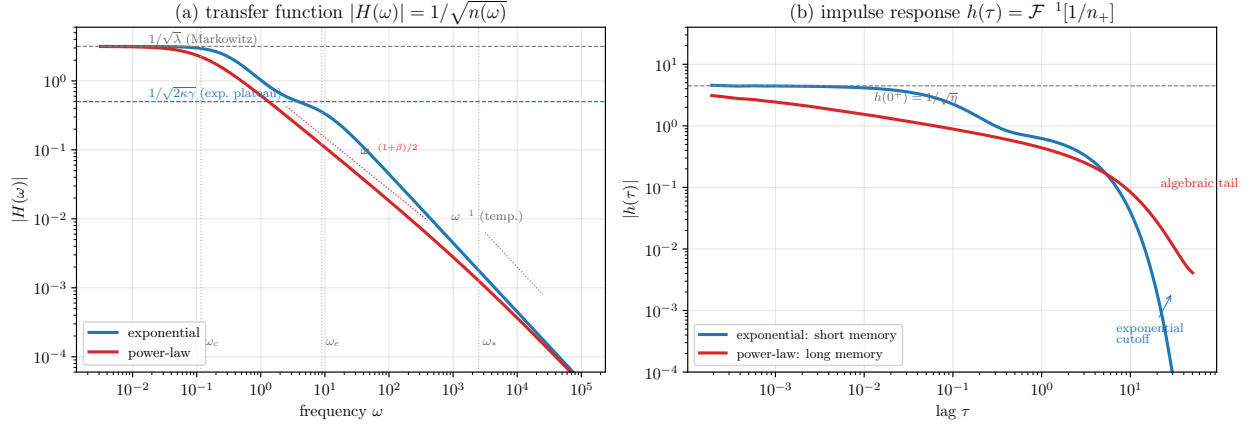


Figure 2: Anatomy of the trading filter, exponential versus power-law transient impact ($\gamma = 1$, $\kappa = 2$, $\beta = 0.5$; small $\lambda = 0.1$, $\eta = 0.05$). (a) Transfer function $|H(\omega)| = 1/\sqrt{\hat{n}(\omega)}$: both share the low-frequency Markowitz plateau $1/\sqrt{\lambda}$ (edge ω_c) and roll off as ω^{-1} under the temporary cost at high frequency (edges ω_e for the exponential, ω_* for the power law); between them the finite-memory exponential settles on the plateau $1/\sqrt{2\kappa\gamma}$ while the long-memory power law slides as $\omega^{-(1+\beta)/2}$. (b) Impulse response $h(\tau) = \mathcal{F}^{-1}[\hat{n}_+^{-1}]$: the temporary cost caps the short lag at $1/\sqrt{\eta}$; past it the exponential filter cuts off (short memory) and the power law leaves an algebraic tail (long memory).

with c_1 from (21), while the position response $X(\theta) = \theta/\Phi(\theta)^2 > 0$ for every kernel. Positions never reverse; the rate reverses exactly when $2c_1\Phi(\theta) > 1$. For the exponential propagator with no temporary cost, $\Phi = \sqrt{A}(m + \theta)/(\kappa + \theta)$ with $A = 2\kappa\gamma + \lambda$ and $m = \kappa\sqrt{\lambda/A}$, and the rate reverses above the threshold

$$\theta^* = \kappa - 2m = \kappa \left[1 - 2\sqrt{\lambda/(2\kappa\gamma + \lambda)} \right]. \quad (23)$$

For the power-law kernel, or any temporary cost, $c_1 = 0$ and $R(\theta) = \theta^2/\Phi(\theta)^2 > 0$ for every θ .

The proof is in Appendix C. When the signal mean-reverts faster than the impact decays, $\theta > \kappa$ at $\lambda = 0$, a signal-following trade would leave an impact residual that outlives the signal, and the optimal rate instead runs against the signal to ride that residual — *impact surfing*. Risk aversion lowers the threshold, and for $\lambda \geq 2\kappa\gamma/3$ the optimal rate runs against the signal at every speed: the position stays aligned with the signal, but its mean reversion dominates the conditional rate. The pure-risk limit $R = -\theta^2/\lambda$ is the extreme case, a Markowitz position always conditionally exiting. For the power-law kernel the rate and position are aligned at every speed and risk aversion; the reversal does not occur there. A temporary cost removes the reversal as well: for $\eta > 0$ the atom vanishes and, at fixed return variance, the value falls with speed as θ^{-2} (§2.4), the per-trade cost of fast trading. Figure 3 shows the rate response and the resulting phase diagram.

The vanishing atom is one face of a broader regularity: the power-law optimum never trades in blocks. A block trade — a jump in the position — pays a transient-impact cost proportional to the kernel at zero lag, $g(0)$, infinite for $g(t) = |t|^{-\beta}$. Instantaneous trading is therefore never optimal, neither on the whole line, where $c_1 = 0$ removes the lag-zero atom, nor at the ends of a finite horizon, where the optimal rate diverges only as the integrable $d(t)^{-\nu}$ in the distance to the endpoint (Proposition 2), so the position stays continuous. The exponential kernel is the

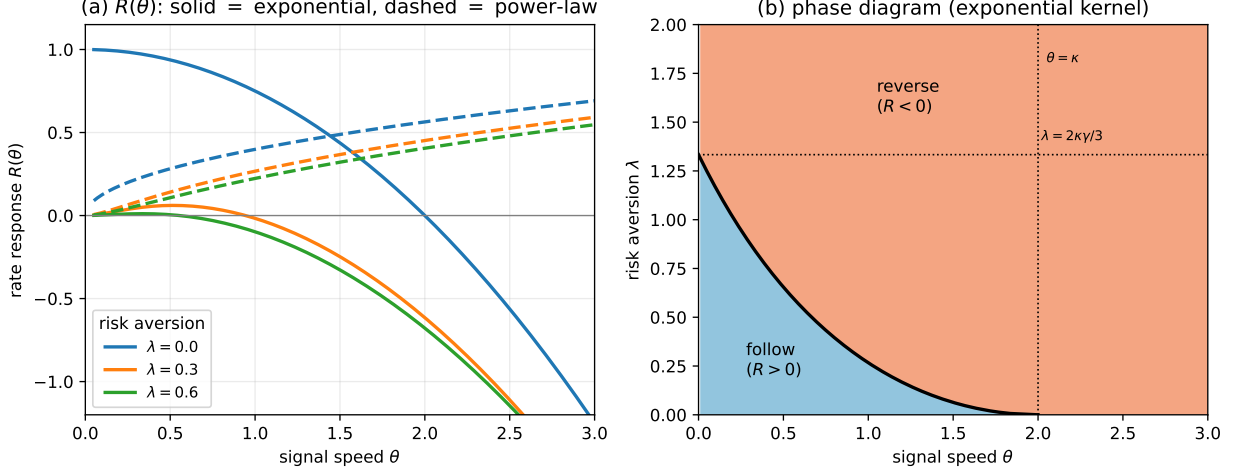


Figure 3: Impact surfing (exponential kernel, $\gamma = 1$, $\kappa = 2$). (a) The rate response $R(\theta)$: the exponential kernel (solid) reverses sign at θ^* as inventory risk λ grows, while the power-law kernel (dashed) stays positive at every risk aversion. (b) Phase diagram: the optimum follows the signal ($R > 0$) below the boundary $\theta^*(\lambda)$ and trades against it ($R < 0$) above; the boundary meets the axes at $\theta = \kappa$ and $\lambda = 2\kappa\gamma/3$.

opposite: $g(0)$ is finite, a block costs only finitely, and the optimal liquidation places discrete trades at both ends — the block-plus-continuous strategy of Obizhaeva and Wang (2013). These distinctions belong to the order-book resilience literature: Alfonsi et al. (2012) treat together whether an optimal execution trades in blocks, whether it stays one-signed (their *positive portfolio problem*), and whether the model admits price manipulation. The first two separate the kernels — the power law, with $g(0) = \infty$, trades in no blocks and, by the positive rate response (22), never reverses sign, while the exponential does both — and the third is ruled out for either by the positive-definiteness $\hat{g} \geq 0$, which also makes impact surfing a cost-optimal response rather than a manipulation (Gatheral 2010, Alfonsi et al. 2012). The frictions omitted here — spread, position limits, order discreteness — would temper the reversal, so the prediction holds within the model class.

4 The finite-horizon factorization

The stationary theory describes an ongoing operation on the whole line. Most execution problems are posed instead on a fixed window $[0, T]$: a broker works a parent order over a session, or liquidates an inherited position by a deadline, ending flat, $x_T = 0$. Two features are then absent from the stationary theory: the two endpoints of the window, and the terminal inventory constraint. On $[0, T]$ translation invariance is lost, the signal and friction no longer share a Fourier basis, and the Szegő formula does not apply.

The factorization survives; only its computation changes. On the window the working variable is again the rate, which carries the terminal constraint and the endpoint blocks; the rate-referred friction on $[0, T]$, $G_T = \eta I$ plus a continuous-kernel integral operator, is positive. Relative to the nest of adapted subspaces such an operator factors triangularly, $G_T = C_- C_+$ with C_+ a causal Volterra operator and $C_- = C_+^*$ (Gohberg and Krein 1970, Arveson 1975). Lemma 1 applies

verbatim, and the optimum is

$$u^\star = C_+^{-1} P_+ C_-^{-1} \alpha^{\text{eff}}, \quad \alpha^{\text{eff}} = \alpha + \sum_k \xi_k e_k, \quad (24)$$

where α^{eff} adjoins a multiplier ξ_k for each linear position constraint $\langle e_k, x \rangle = 0$, with a process-valued multiplier for the pathwise liquidation constraint $x_T = 0$; each modifies the signal exactly as the adaptedness constraint does in (10), confined to its constraint's annihilator, as in the constrained propagator problem of Abi Jaber et al. (2024a). The Szegő formula is replaced by a Volterra kernel, and nothing else in the construction moves.

4.1 Relaxation to the stationary filter

The two endpoints and the terminal constraint act locally: their influence decays into the interior, where the finite-horizon policy relaxes to the stationary filter of §2.3. The bulk of a long session is governed by the whole-line solution, and only the start-up and terminal windows need separate treatment.

Proposition 2 (Boundary-layer decay). *Let $u^{\star,T}$ be the finite-horizon optimum on $[0, T]$, $u^\star$ the whole-line stationary optimum for the same bounded signal, and $d(t) = \min(t, T - t)$ the distance to the nearest endpoint. Then, for constants $C(\beta)$, C depending only on the kernel parameters,*

$$|u_t^{\star,T} - u_t^\star| \leq \begin{cases} C(\beta) \|\alpha\|_\infty d(t)^{-\nu}, & \text{power-law kernel, } \nu = \frac{1-\beta}{2}, \\ C \|\alpha\|_\infty e^{-b_1 d(t)}, & \text{rational symbol, } b_1 \text{ the slowest zero of } \hat{n}_+. \end{cases} \quad (25)$$

The proof is in Appendix D. The boundary layer has width $\sim 1/b_1$ for a rational kernel and is set by the impact memory for a power law; it is thin except for the small ν of the empirical power-law exponents, a quantitative statement of how long-memory impact makes horizons matter. Figure 4 shows the finite-horizon rate tracking the stationary filter through the interior.

4.2 The power-law factor

For a scale-free kernel the Volterra factor is explicit. For the power-law kernel ($\eta = \lambda = 0$) the causal factor C_+ is the terminal-anchored Volterra operator with kernel

$$c_+(t, s) = (\gamma c_\beta)^{1/2} \left(\frac{T-s}{T-t} \right)^\nu \frac{(t-s)^{\nu-1}}{\Gamma(\nu)}, \quad 0 \leq s \leq t \leq T, \quad (26)$$

verified by direct kernel integration, $C_- C_+ = G_T$ including the constant. This is the whole-line fractional factor I_+^ν conjugated by multiplication by $(T-t)^\nu$: the terminal weight anchors the convolution factor to the endpoint T , and homogeneity of the kernel is what allows the anchoring to be a scalar weight. Each factor sees one endpoint, C_+ the terminal and its time reflection C_- the initial; the identity (8) places the causal factor on the right of (24), so the adapted projection sits between the factors and forces C_+ to anchor at T . Away from the endpoints the weight tends to one and C_+ tends to the whole-line factor, which is Proposition 2 for this kernel.

4.3 General kernels

Between the scale-free power law and the rational symbols the factor is not explicit. The factorization $G_T = C_- C_+$ still exists (Gohberg and Krein 1970, Arveson 1975), but the factor kernel

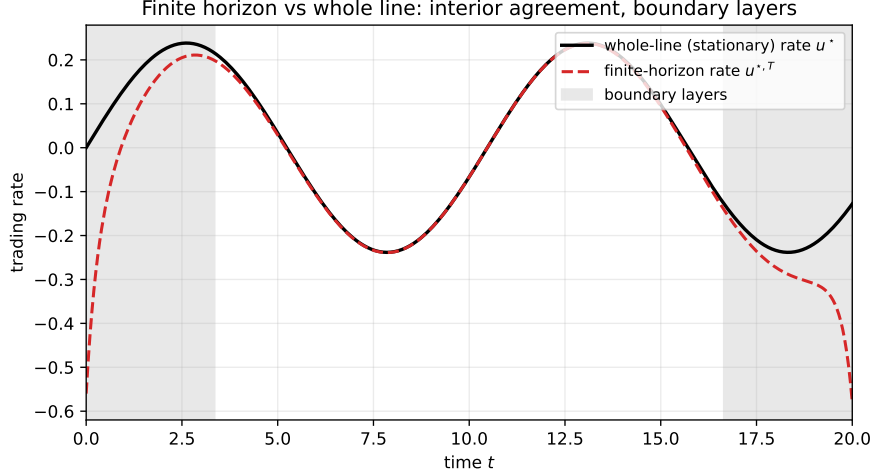


Figure 4: Finite horizon versus whole line ($\eta = 0.5$, $\gamma = 1$, $\kappa = 2$, $\lambda = 1$, $T = 20$). The finite-horizon optimal rate tracks the whole-line stationary rate through the interior and departs only in the start-up and terminal boundary layers, as quantified by Proposition 2; the shading marks the boundary-layer scale, not a fitted width.

solves the Gohberg–Krein integral equations on $[0, T]$, with a closed form only in two classes: rational symbols, where it is a finite sum of endpoint-anchored exponential modes matched at the two boundaries, and scale-free kernels, where it is the conjugated fractional factor (26). For a general propagator the equations are solved numerically.

5 Recovery of earlier solutions

The portfolio and execution rules of the literature are special cases of the factorization, in two groups. When the friction is stationary and rational — pure risk, temporary cost, exponential resilience, and their combinations — the Szegő factor is a ratio of polynomials, and the optimal position is a finite sum of exponential moving averages of the signal (Section 2.3); Markowitz, the Gârleanu–Pedersen aim portfolio, and the stationary Neuman–Voß solution are the cases with zero, one, and two averages. When the horizon is finite, the Gohberg–Krein factor of Section 4.2 takes over, producing the liquidation profiles of the execution literature — the power-law U-shape and Fredholm solutions, and the exponential block-plus-continuous strategy of Obizhaeva–Wang. Beyond these two, a general propagator has no closed factor; §5.3 identifies the factorization with the general-propagator solutions of the literature. Figure 5 shows how the shape of the policy is set by the impact memory across this family.

5.1 Stationary filters: rational frictions

For a rational symbol the optimal position is a finite sum of exponential moving averages of the return signal: their rates are the zeros of $\hat{n}_+$, their number is the number of poles, and risk aversion sets their scale.

Risk alone: the Markowitz rule. With $\eta = \gamma = 0$ the symbol is the constant $\hat{n} \equiv \lambda$, the factorization is trivial, and the *position* is the memoryless mean–variance rule $x_t^* = \mu_t/\lambda = \mathbb{E}_t[-\dot{\alpha}_t]/\lambda$ of Markowitz (1952), Merton (1971), the expected return over risk aversion. The position responds to

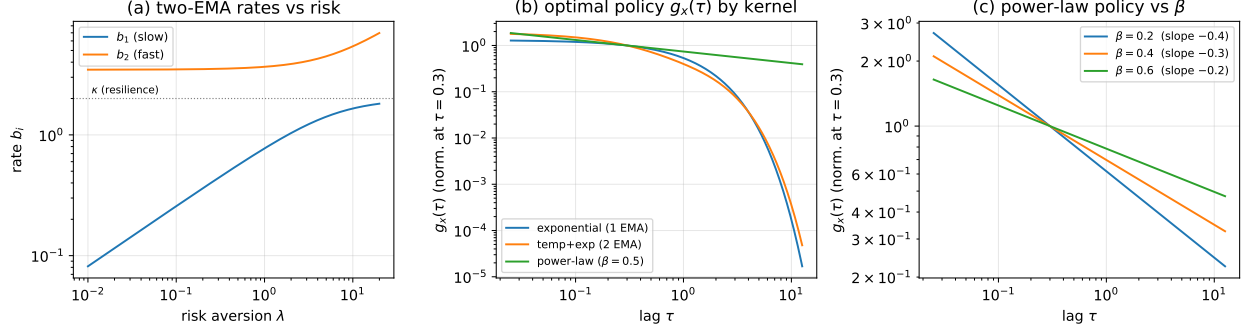


Figure 5: The optimal policy across kernels and parameters. (a) For the temporary-plus-resilient kernel ($\eta = 0.5$, $\gamma = 1$, $\kappa = 2$), the two moving-average rates b_1, b_2 against risk aversion: the slow rate $b_1 \rightarrow 0$ as $\lambda \rightarrow 0$ (the position becomes non-stationary without risk) and both grow with λ . (b) The position impulse response $g_x(\tau)$ by kernel ($\theta = 1$, normalized at $\tau = 0.3$): the exponential and two-average kernels have finite memory (exponential cutoff), the power-law kernel a heavy tail, and the Markowitz limit is an instantaneous spike. (c) The power-law policy across β : a scale-free kernel of slope $(\beta - 1)/2$.

the return μ , whereas under temporary impact alone (Section 3) the trading *rate* is proportional to the appreciation signal α . For an Ornstein–Uhlenbeck signal $\mu_t = \theta\alpha_t$, and the value is $v = \theta\sigma^2/4\lambda$.

Temporary cost with risk: the aim portfolio. With $\gamma = 0$ the friction is the differential operator $N = \lambda - \eta\partial_t^2$, local like the Markowitz case, $\hat{n}_+ = \sqrt{\eta}(a - i\omega)$ with $a = \sqrt{\lambda/\eta}$: one moving average, and differentiation puts the filter in partial-adjustment form,

$$u_t^* = a(\text{aim}_t - x_t^*), \quad \text{aim}_t = \frac{a}{a + \theta} \cdot \frac{\theta\alpha_t}{\lambda} : \quad (27)$$

trade at speed a toward an aim that discounts the Markowitz position by the persistence factor $a/(a + \theta)$. This is the continuous-time aim portfolio of Gârleanu and Pedersen (2016) (discrete time, Gârleanu and Pedersen (2013); general targets, Bank et al. (2017)): a diagonal friction gives exactly one pole.

Exponential propagator with risk. For $g = e^{-\kappa|t|}$ and $\eta = 0$, $\hat{n}_+ = \sqrt{A}(m - i\omega)/(\kappa - i\omega)$ with $A = 2\kappa\gamma + \lambda$, $m = \kappa\sqrt{\lambda/A}$, and (15) becomes the current signal plus one moving average,

$$x_t^* = \frac{\kappa + \theta}{A(m + \theta)} \left[\mu_t + (\kappa - m) \int_{-\infty}^t e^{-m(t-s)} \mu_s ds \right], \quad (28)$$

whose rate m travels from 0 to κ as risk aversion grows. Exponential resilience with an Ornstein–Uhlenbeck signal is the model of Lehalle and Neuman (2019), whose explicit solution is the risk-free, singular finite-horizon liquidation generalizing Obizhaeva and Wang (2013); (28) is its stationary, no-temporary-cost form, risk aversion holding the position in place of the horizon.

Exponential propagator with temporary cost and risk. With all three frictions the symbol is rational, and the outer factor $\hat{n}_+ = \sqrt{\eta}(b_1 - i\omega)(b_2 - i\omega)/(\kappa - i\omega)$ has two zeros ($b_1^2 b_2^2 = \lambda\kappa^2/\eta$, $b_1^2 + b_2^2 = \kappa^2 + (2\kappa\gamma + \lambda)/\eta$), so (15) becomes a two-average filter, $x_t^* = \frac{1}{\Phi(\theta)\sqrt{\eta}} \sum_i w_i \int_{-\infty}^t e^{-b_i(t-s)} \mu_s ds$ with $w_i = (\kappa - b_i)/(b_j - b_i)$. Neuman and Voß (2022) solve this by a system of coupled forward–backward SDEs and give the strategy as a feedback control; its gains reduce, in the stationary regime, to these two rates and two weights — the zeros and residues of the symbol — and the finite-horizon gains converge to them in the interior, the closed-loop poles equalling the moving-average rates b_1, b_2 . Both degenerations are exact: at $\gamma = 0$ the rates are $\{a, \kappa\}$ and the κ -pole

cancels the kernel zero, recovering (27); as $\eta \rightarrow 0$, $b_1 \rightarrow m$ and $\sqrt{\eta}(b_2 - i\omega) \rightarrow \sqrt{A}$, recovering (28). A rational friction with n poles gives a filter with at most n averages.

5.2 Finite-horizon liquidation profiles

On a finite horizon the liquidation profiles of the execution literature follow from (24). For a power-law kernel and a constant signal the projection is trivial and (24) reduces to $G_T^{-1}\mathbf{1}$ up to the terminal-inventory multiplier, the U-shaped profile $u^*(t) \propto [t(T-t)]^{(\beta-1)/2}$ of Gatheral et al. (2012, Ex. 2.30). The exponential kernel gives the rational counterpart: a discrete block at each endpoint and continuous trading between, the block-plus-continuous liquidation of Obizhaeva and Wang (2013), its two blocks the finite-horizon form of the white-noise atom of §3.2. For a Gaussian Volterra signal, the first-order condition of the finite-horizon problem is the stochastic Fredholm equation that Forde et al. (2022) solve on the Wiener chaos; the time reflection of (26) is the left-anchored outer factor they use to invert G_T in full, where the factor order is immaterial. Formula (24) is its solution operator, and expanding it on the chaos reproduces their Fredholm systems, with the generalized Abel inversion of Chakrabarti and George (1994) implementing G_T^{-1} and weighted Riemann–Liouville operators the individual factors.

5.3 General propagators

A general propagator has no closed factor (§4.3), so the explicit rules above do not apply, but the factorization still represents the solution. The adapted optimum of (1) is unique (Theorem 1), so the Gohberg–Krein policy (24) and the general-propagator solutions of the literature are one process written several ways: the free-boundary backward SDE and operator Riccati equation of Abi Jaber and Neuman (2025), the operator resolvents of Abi Jaber et al. (2024b), and the stochastic Fredholm equation with constraint multipliers of Abi Jaber et al. (2024a). On $[0, T]$ the identification is term by term: the iterative solution of the stochastic Fredholm equation is the Neumann expansion that the Gohberg–Krein factors sum in closed form. In the stationary regime the Szegő factor gives the closed-form filter of §2.3, and in the interior the finite-horizon solution relaxes to it (Proposition 2, proved for the rational and power-law kernels and expected more generally).

6 Concluding remarks

Trading against impact is a problem of timing. Unconstrained, the first-order condition is a single inversion of the friction operator; under adaptedness, with a friction that couples different times, the trader cannot truncate that inversion, because projection onto the past does not commute with the inverse. Factoring the friction along the filtration, into a causal part and its anticausal adjoint, resolves this, and the adapted inverse assembles from the one-sided factors — whiten the forecast, project, recolour.

On the whole line the factor is a frequency-domain formula and the policy is a *filter*, a transfer function that maps the observed signal to the position; the classical portfolio and execution rules are its special cases — Markowitz when the friction is constant, the aim portfolio when it is a second-order differential operator, the moving-average filters of the resilience models, and a fractional derivative when the impact is scale-free. The shape of the policy is set by the impact memory: a rational kernel has finitely many relaxation rates and the position is a finite sum of moving averages; a power-law kernel has a continuum and the policy is a fractional derivative, keeping a fraction $\sin(\pi\beta/2)$ of the value a clairvoyant trader would earn.

Three restrictions set the scope. The costs are quadratic in the rate and position, so nonlinear impact lies outside the model. The explicit filters require a stationary Gaussian signal; for a general adapted signal, Theorem 1 gives the policy through the forecast curve but not as a transfer function. And the asset is one-dimensional: a portfolio version calls for a matrix Wiener–Hopf factorization, explicit only in special cases. On finite horizons the theory is complete up to the boundary terms attached to constraints, for which (24) gives the operator form but not yet sharp constants.

A Proof of Lemma 1 and Theorem 1

Causality of A_+ gives $(I - P_+)A_+P_+ = 0$, so A_+ and its causal inverse preserve the adapted subspace; taking adjoints and using $A_- = A_+^*$ gives $P_+A_-(I - P_+) = 0$, so A_- and A_-^{-1} preserve the orthogonal complement. For adapted v , write $A_-v = P_+A_-v + (I - P_+)A_-v$; applying A_-^{-1} , the image of the second term stays in the complement and is annihilated by P_+ , so $P_+A_-^{-1}P_+A_-v = P_+v = v$. Hence for adapted u ,

$$A_+^{-1}P_+A_-^{-1}(P_+AP_+)u = A_+^{-1}[P_+A_-^{-1}P_+A_-]A_+u = A_+^{-1}A_+u = u,$$

using $P_+A_+u = A_+u$. The reverse composition equals the identity by the same bookkeeping, or from positivity of P_+AP_+ on the adapted subspace, for which a one-sided inverse is two-sided. For the pure power-law kernel the factors are unbounded and the computation holds on the dense domain of signals with $\int (1 + \omega^2)S_\alpha(\omega)/\hat{q}(\omega) d\omega < \infty$ (Appendix B); any $\eta > 0$ or $\lambda > 0$ gives bounded inverses.

Theorem 1 follows with $A = N$: strict convexity ($\hat{n} > 0$ almost everywhere) makes (4) necessary and sufficient with a unique solution, which Lemma 1 inverts. The forecast-curve form follows from conditional Fubini: N_-^{-1} has a deterministic anticausal kernel, so $\mathbb{E}_s[(N_-^{-1}\mu)(s)]$ integrates that kernel against $\mathbb{E}_s[\mu_r] = \bar{\mu}(s, r)$ for $r \geq s$. The rate form (18) follows by differentiating (9): $\partial_t \circ N_+^{-1} = Q_+^{-1}$, and on forecast curves $Q_-^{-1} = N_-^{-1} \circ (-\partial_s)$ with $-\partial_s \bar{\alpha}(s, \cdot) = \bar{\mu}(s, \cdot)$, so the whitened forecasts coincide.

B Proof of Theorem 2 and the Ornstein–Uhlenbeck formulas

The whitened return $N_-^{-1}\mu$ is the stationary filter of $\dot{W}$ with symbol $h = \hat{\psi}\hat{n}_-^{-1}$. Conditional expectation at s annihilates innovations after s , so P_+ truncates the moving-average kernel at lag zero, acting as the plus-part $[\cdot]_+$ on the symbol. Composing with N_+^{-1} gives (11); for $\lambda > 0$ the symbol $\hat{n}$ is bounded below, so h and the position filter lie in L^2 . The hypothesis $\int (1 + \omega^2)S_\mu(\omega)/\hat{n}(\omega) d\omega < \infty$ places h in a Sobolev class on which the half-line indicator multiplies boundedly (Lions and Magenes 1972, Ch. 1), giving $\hat{\chi} \in L^2$. For the value, the first-order condition and adaptedness give $v = \frac{1}{2}\mathbb{E}[x_t^*\mu_t]$, and by the Itô isometry with $\hat{\psi} = \hat{n}_+\bar{h}$ on the real axis, $\mathbb{E}[x^*\mu] = \frac{1}{2\pi}\| [h]_+ \|^2$, which is (12).

Referred to the rate, the same argument with $A = Q$ gives $\hat{\chi} = \hat{q}_+^{-1}[h_\alpha]_+$ and $v = \frac{1}{4\pi}\| [h_\alpha]_+ \|^2$ with $h_\alpha = \hat{q}_+^{-1}\hat{\varphi}$, valid whenever $h_\alpha \in L^2$, the power law at $\lambda = 0$ included. The two forms agree where both apply, since $\hat{\psi} = \varphi(0) + i\omega\hat{\varphi}$ makes $h - h_\alpha$ anticausal up to a lag-zero atom that the plus-part annihilates.

For the Ornstein–Uhlenbeck signal, $\hat{\psi} = \sigma\theta/(\theta - i\omega)$ and $h = \hat{\psi}\hat{n}_-^{-1}$; the combination

$$h - \frac{\hat{\psi}}{\Phi(\theta)} = \sigma\theta \left[\frac{1}{\hat{n}_-(\omega)} - \frac{1}{\hat{n}_-(-i\theta)} \right] \frac{1}{\theta - i\omega}$$

is anticausal — the bracket’s pole at $\omega = -i\theta$ cancels, $\hat{n}_-^{-1}$ is analytic in the lower half-plane, and $\hat{n}_-(-i\theta) = \hat{n}_+(i\theta) = \Phi(\theta)$ by conjugate symmetry. Hence $[h]_+ = \hat{\psi}/\Phi$, giving (15) and, with $\|\hat{\psi}\|^2 = \pi\sigma^2\theta$, the value $v = \sigma^2\theta/4\Phi^2$. For the power law at $\lambda = 0$, $\hat{q}_- = (\gamma c_\beta)^{1/2}(-i\omega)^{-\nu}$ gives $\hat{n}_-(-i\theta) \propto \theta^{(1+\beta)/2}$ and, via (16), the ratio $v/v_{\text{ant}} = \sin(\pi\beta/2)$ of (17), the sine arising from the argument of $(-i\omega)^{\beta-1}$ on the two half-lines.

C Proof of Proposition 1

Write $\hat{\varphi} = \sigma/(\theta - i\omega)$ for the outer factor of $\alpha = \mu/\theta$, so the position filter is $(\theta/\Phi)\hat{n}_+^{-1}\hat{\varphi}$. It vanishes at high frequency for every kernel considered, so $X(\theta)$ is the Laplace evaluation $\theta/\Phi(\theta)^2 > 0$. The rate filter $\hat{\chi} = (-i\omega)(\theta/\Phi)\hat{n}_+^{-1}\hat{\varphi}$ tends to $a_0 = \theta\sigma c_1/\Phi$; splitting $\chi = a_0\delta + \chi_{\text{reg}}$, the atom is excluded from the forward limit, and dominated convergence gives $R = (2\theta/\sigma^2)\sigma[\hat{\chi}(i\theta) - a_0] = \theta^2\Phi^{-1}[\Phi^{-1} - 2c_1]$. For the exponential kernel with $\eta = 0$, $\hat{g}(\omega)\omega^2 \rightarrow 2\kappa$, so $\hat{n}(\infty) = \lambda + 2\gamma\kappa = c_1^{-2}$; the power-law kernel and any $\eta > 0$ make $\hat{n}_+$ grow, so $c_1 = 0$. The threshold (23) follows from $\Phi = \sqrt{A(m+\theta)/(\kappa+\theta)}$; consistency checks: $\lambda \rightarrow 0$ on the exponential returns $R = (\kappa^2 - \theta^2)/2\kappa\gamma$, and $\gamma \rightarrow 0$ returns $R = -\theta^2/\lambda$.

D Proof of Proposition 2

The finite-horizon policy (24) differs from the stationary one through the terminal-anchored weight and the truncation of the factor kernels to $[0, T]$. For the power-law factor (26) the truncation cuts the Marchaud tail at distance $d(t)$, contributing $d(t)^{-\nu}$; the weight deviation $((T-s)/(T-t))^\nu = 1 + O(d(t)^{-1})$ is subdominant. For a rational symbol the causal factor $\hat{n}_+^{-1}$ has kernel decaying as $e^{-b_1\tau}$ at the slowest zero b_1 of $\hat{n}_+$, so truncation at distance $d(t)$ costs $e^{-b_1d(t)}$; equivalently the finite-horizon feedback gains relax to their stationary values at rate $2b_1$.

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